



ME492 Final Execution Plan

Individual execution baseline | Revision 1.1 | 30 September 2026

Individual ME492 execution baseline: a reproducible engineering study of one Arche Robotics mobile robot. Robot access, CAD, telemetry and existing AI model access remain company decisions.

Primary and secondary objectives

Primary objective: develop and evaluate a caster-aware digital twin and compare conventional and smoother motion profiles. Secondary objective: assess whether the same experimental data supports a compact short-horizon motion-error predictor.

Research questions

RQ1: To what extent does an explicit caster-aware model improve the reproduction of measured mobile-robot motion compared with an ideal drive model?

RQ2: Can a smoother speed/turn profile reduce tracking/yaw error without an unacceptable increase in completion time?

RQ3: Can recent telemetry predict short-horizon yaw tracking error better than simple persistence/rule baselines?

Defined semester scope

Deliverable	Bounded commitment
Robot and mechanics	One configuration; measured/supplied geometry, ideal-drive baseline and explicit caster swivel/rolling dynamics.
Simulation and comparison	Two approved surfaces; straight/braking, left/right circle, stop-turn-go and reversal; one path tracker.
Optimization	Small caster/contact sensitivity study; one smoother speed/turn profile. Hardware changes need separate permission.
Learning model - secondary	One numerical target, simple baselines and one compact model; dataset sufficiency and independent test results.
Final package	English report/defense, permitted files/logs, manifests, parameters, analysis and reproduction instructions.

Fixed academic milestones

GDS: 6 November, 12:00–15:00. Draft: 4 December. Final report: 13 January 2027, 23:59. Defense: 14 January, published session 09:30–12:00. Calendar checked 30 September. [1]

30 September–14 January: 252 planned hours plus 32 contingency hours; reserve is not feature development. Target Arche visit: Tuesday 6 or Wednesday 7 October, unconfirmed. Previous research architecture remains the technical reference. [3]



Capacity and Acceptance Criteria

Planning estimates and evidence requirements

This schedule assumes roughly 18–20 project hours in ordinary weeks, with lower loads around GDS and finals. AI assistance supports coding and writing, but does not reduce robot setup, measurement, debugging or the obligation to explain the work. At a sustained 15 hours per week, the full scope is unlikely to fit without reducing the learning-model work or experiment breadth.

Work package	Planned hours
Scope, company coordination and protocol	25
ROS and simulator learning on project tasks	42
CAD, mechanics, twin and identification	65
Physical experiments and data processing	38
Trajectory optimization and comparison	24
Learning task, baselines and evaluation	28
Report, reproducible handoff and defense	30
Total planned work	252
Contingency held outside task estimates	32
Total capacity required	284

The weekly hours include learning, visits, ordinary local travel, meetings and writing. Company waiting time is a calendar dependency, not productive work. Reserve 24 contingency hours for access, logging and model repairs before 18 December and 8 for final corrections. Do not use this reserve to add features. Weekly availability and long travel times are reviewed after the first visit.

Completion tests

RQ1: compare ideal-drive and caster-aware predictions under identical measured commands on held-out runs. Report pose/yaw error per run, parameter sources and contact/time-step sanity checks. Physical claims require independent pose and timestamp checks. Geometry-only and uncalibrated models remain labelled.

RQ2: compare tracking/yaw error, acceleration or jerk, and completion time under matched routes and limits. A proposed 10% relative error reduction with at most 10% extra time is a planning target to agree at G1, not a predicted result or graduation pass rule. Lock the time penalty and evaluation settings before confirmation trials.

RQ3, the secondary objective: document inputs/targets, whole-run splits, baselines, feasible weights and held-out prediction MAE. No improvement is a valid finding. Numerical accuracy, model usefulness and physical deployment remain separate outcomes. I own implementation, experiments, analysis and reporting; the advisor supervises academically and company engineering labor is not assumed.



Timetable September to November

Owner Oluş Emre Demir Planned work 30 September to 22 November

Dates describe the final execution baseline. Company visits and robot sessions are requests until Arche confirms them. GDS is 6 November. Draft report is 4 December. Final report is 13 January, 23:59, followed by the 14 January presentation. [1]

Period	Hours	Work and output	Acceptance evidence
30 Sep–4 Oct	10	Scope and preparation. Consolidate the existing research report. Define hypotheses, four maneuvers and required channels. Prepare the visit and verify the available development computer.	Access checklist, experiment outline and a reproducible ideal-drive calculation.
5–11 Oct	18	Arche technical visit. Target Tue 6 Oct or Wed 7 Oct. Inspect the actual platform, software and data. Select the compatible simulation environment. Record access decisions.	Visit record with owners and due dates. Nominal robot dimensions and a sample telemetry file requested.
12–18 Oct	18	Platform and ideal model. Measure or import the chassis, drive wheels and caster geometry. Implement the ideal differential-drive baseline. Agree pose-reference and logging methods.	G1 on 16 Oct. Correct straight/circle motion, explicit units and frames, agreed physical access route.
19–25 Oct	18	Caster model and physical pilot. Add passive swivel/rolling joints and a small friction/damping model. Run a supervised logging pilot on Tue 20 Oct or Wed 21 Oct.	G2 on 23 Oct. One usable real log with independent pose where permitted. Stable caster reversal simulation.
26 Oct–1 Nov	16	Repeatable benchmark. Support two surface patches and four maneuvers. Replay identical commands. Check time alignment, sensor gaps and contact-step sensitivity. Allow for the 28–29 Oct holiday.	Versioned run manifest, identical command replay and automated metric export. Reference protocol frozen.
2–8 Nov	10	Protocol review and GDS. Fix logging defects and freeze calibration/test separation. Keep the week light for GDS on Fri 6 Nov, 12:00–15:00.	G3 on 6 Nov. Valid data route, pose-reference quality and approved test slots. No new major subsystem.
9–15 Nov	20	Calibration experiments. Target Tue 10 Nov or Wed 11 Nov for 24 nominal-profile runs. Fit measured geometry, drive delay and a small caster/contact parameter set.	Calibration dataset and data-quality report. Parameter bounds, fitted values and simple-model comparison.
16–22 Nov	20	Trajectory optimization. Evaluate caster/contact sensitivity in simulation. Select one smoother speed/turn profile using the same path tracker. Reserve untouched physical test runs.	G4 on 20 Nov. Twin parameters and finalist profile locked before confirmation trials. First comparative plots.

Subtotal: 130 planned hours. Writing and the experiment log start in the first week. If the first visit moves to 13–14 October, use the preceding week for ideal-drive and logging preparation and keep the access decision deadline at 16 October.



Timetable November to January

Owner Oluş Emre Demir Planned work 23 November to 14 January

University finals run from 26 December to 8 January. [2] Main features and experiments freeze on 18 December. The internal report target is 23 December, package target 25 December and submission-ready target 8 January.

Period	Hours	Work and output	Acceptance evidence
23–29 Nov	18	Confirmation data and AI baseline. Target Tue 24 Nov or Wed 25 Nov for 48 paired baseline/improved-profile runs. Seal final test files. Build causal features and a small regression baseline from development data.	Separate calibration, development and final-test run IDs. Confirmation matrix recorded without tuning on final results.
30 Nov–6 Dec	18	Draft report and model freeze. Complete methods and initial development results. Submit the draft on Fri 4 Dec. Fit one compact learned predictor and lock all evaluation settings by 6 Dec.	Draft report submitted. G5 on 6 Dec. Model artifact or documented data-feasibility result, baselines and frozen metrics.
7–13 Dec	20	Independent evaluation. Open the final holdout once. Compare ideal/caster-aware twins, baseline/improved profiles and persistence/rule/learned predictors. Use Tue 8 Dec or Wed 9 Dec only for a justified retry.	Held-out error, spread, completion time and model usefulness. Negative findings retained. Repeated tuning needs a new test set.
14–20 Dec	20	Interpretation and final analysis. Resolve specific mismatches. Quantify uncertainty and feasible caster recommendations. Prepare the final report, graphs and a short reproducible demonstration.	G6 on 18 Dec. Main experiments and features frozen. Traceable figures, limitations and conclusions.
21–27 Dec	16	Package and advisor review. Deliver an internal final report by 23 Dec and a reproducible package by 25 Dec. Finish pending corrections before finals begin on 26 Dec.	Advisor review copy, data/model inventory, reproduction instructions and final presentation outline.
28 Dec–3 Jan	8	Finals and reserved corrections. Use a reduced project load during university finals. Correct report text and broken reproduction steps. New experiments require a specific unresolved claim.	Reviewed report and checked references. No expansion of the main scope.
4–10 Jan	10	Submission and defense preparation. Prepare the submission PDF, archive and demonstration recording. Rehearse the method, assumptions, graphs and failure cases. Target submission-ready files on 8 Jan.	Final files verified by 8 Jan. Two rehearsals and a backup demonstration ready by 10 Jan.
11–14 Jan	12	Final handoff. Resolve last advisor comments. Submit the final report by 13 Jan, 23:59. Present on 14 Jan within the published 09:30–12:00 session window.	Submission receipt and defense. Individual presentation slot follows the course announcement.

Subtotal: 122 planned hours. Combined timetable: 252 hours. The 32-hour reserve is additional capacity, used only for named problems. Final-test files stay separate from development data until the model and evaluation settings are locked.



Decision Gates and Access Contingencies

Dependencies that determine the permitted conclusions

Gate	Evidence and decision
G1 16 Oct	Platform/software identified, allowed data exports and robot slots recorded. Pose-reference approach agreed. Otherwise freeze a provisional robot model and request the missing items with a due date.
G2 23 Oct	Stable nominal/caster simulations and one usable logging pilot. If CAD is unavailable, use measured simplified geometry and retain that limitation.
G3 6 Nov	Data route, independent pose and experiment access pass quality review. Without them, seek advisor agreement by 9 Nov for the reduced validation scope.
G4 20 Nov	Calibration complete enough to define a defensible comparison. Lock the finalist profile before the confirmation dataset. Simplify unidentifiable parameters.
G5 6 Dec	Model, input/target definition and test protocol frozen. If labels or independent runs are inadequate, deliver a feasibility analysis and rule/numerical baseline instead of an unsupported performance claim.
G6 18 Dec	Main results, figures and scope frozen. Remaining work is correction, packaging and defense.

Physical access routes

Route A is supervised student access for the pilot, calibration and confirmation sessions. Route B uses an Arche operator replaying the same approved commands and exporting logs plus a pose reference. Route B is acceptable only if configurations, timing, repeats and permissions remain traceable.

Historical logs can support identification if commands, configuration and time references are complete. Without an independent pose reference, report observable yaw/encoder/caster comparisons and limit trajectory-accuracy claims. Without any suitable real logs by 6 November, the project becomes a simulation study with a prepared physical protocol, subject to advisor acceptance. Physical validation remains uncompleted.

A delayed visit does not shift the official deadlines. Decline extensions to surfaces, hardware changes, extra models and deployment. If usable capacity still falls below 252 hours, reduce four maneuver families to two with the advisor, retaining repeated comparisons and the report. A necessary reduction of the scheduled learning experiment is also recorded with the advisor.

Critical path

Platform access enables a usable pilot. The pilot enables calibration data. Calibration enables model/profile locking. The independent confirmation set enables evaluation and conclusions. Writing proceeds alongside each step. A learning model cannot remove any of these dependencies.



Physical Experiment Protocol

One platform two surfaces and four maneuver families

Use two accessible, safe patches selected with Arche. Start with a smooth indoor surface and a second contrasting approved surface. Pavers, ramps and extra payloads are extensions after the core runs succeed. Record the exact patch rather than assigning universal friction values to a material name.

Session request	Purpose and target
20 or 21 Oct 2–3 hours	Pilot: check commands, timestamps, frames, pose reference, caster visibility and restart behavior. Pilot runs are separate from the final test.
10 or 11 Nov 3–4 hours	Calibration: 2 surfaces $\times$ 4 maneuver families $\times$ 3 repeats = 24 nominal-profile runs. Use whole-run development splits.
24 or 25 Nov 3–4 hours	Confirmation: 2 surfaces $\times$ 4 maneuver families $\times$ 2 profiles $\times$ 3 repeats = 48 runs. Balance baseline/improved order and seal final-test files.
8 or 9 Dec reserve 2–3 hours	Repeat only defective recordings or a specific unresolved comparison. If results guide tuning, collect a new untouched test set.

These slots total 8–11 hours before the optional retry; all remain requests. Seventy-two main runs are a planning target. Three repeats per cell are an initial floor, not a statistical power guarantee: they are the minimum practical replication under the requested robot-access budget. Report uncertainty per run/cell rather than treating the repetitions as a powered population study. Typical runs take 30–90 seconds plus reset/checks. If resets exceed the slot budget, agree a reduction to two maneuver families before final-test collection.

Measurements and controls

Record issued motion commands, wheel states/encoders, IMU, odometry, independent pose, source timestamps and logger timestamps where available. Prefer direct caster angle from an approved marker/video or sensor. Motor current is useful when available. Log geometry/configuration revision, payload, surface, battery state, operator and starting pose/caster angle for every run.

Validate reference scale and orientation against a known distance and a stationary test. Set sample rates from observed dynamics during the pilot. No shimmy claim follows from a signal whose sampling cannot resolve the oscillation. Preserve raw files and flags for gaps, resets and clock drift. Never use the odometry under evaluation as its own pose ground truth.

Arche sets safe speed, acceleration, area and stop rules. A designated operator supervises physical trials and can stop motion. I change only approved commands or profiles. Platform safety and low-level motor control remain under the existing system.



Learning Model Work Package

Secondary objective | Motion prediction within 28 hours

Prediction task

RQ3, a secondary objective, reuses RQ1/RQ2 recordings to test predictive value beyond simple baselines. Predict yaw tracking-error magnitude 0.5 seconds ahead from roughly 0.5 seconds of causal commands, speed, yaw rate, acceleration and recent error. Include caster angle/current only if reliable. Freeze the horizon and channels on development data. This is a motion-disturbance proxy; caster attribution requires caster measurements.

Smallest defensible model

Compare persistence, a transparent numerical/rule baseline and one regularized regression. A compact nonlinear model requires a specific development gap and room in the 28-hour budget. Check any existing Arche model for task fit and permission before reuse. A language model is not the default for numerical telemetry. Foundation-model training, RL and a full robot intelligence stack are outside scope.

Step	Hours	Output
Task, labels and run split	6	Causal feature/target specification and provenance.
Data checks and simple baselines	8	Persistence/rule results and leakage checks.
Training and development comparison	8	One model, preprocessing and locked settings.
Independent evaluation and write-up	6	Test metrics, model card and benefit/limitation decision.

Evaluation rules

Split by entire run and preferably by recording session. Fit scaling and feature selection on training data only. The 24 calibration runs provide development data; blocked validation reserves whole runs. The 48 confirmation runs remain untouched until 7 December. Overlapping windows are correlated and do not create independent robot trials. Synthetic data remain labelled and cannot constitute real-world validation.

Report prediction MAE against persistence, per-run spread and performance across the two surfaces. If a warning threshold is added, choose it on development data and report missed events and false warnings per minute on the holdout. Rare events or inadequate labels trigger G5's feasibility outcome. Report inference latency on the actual target computer; laptop timing alone cannot establish robot deployment readiness.

The model can remain offline. Deployment requires runtime-available channels: external reference error can label evaluation but cannot become an input without that reference online. Deployment needs Arche authorization and held-out benefit. A second platform or broader events belong to later work supported by evidence and permission.



Arche Robotics Technical Visit

Platform access experiment feasibility and model interfaces

Visit theme: enabling a measurable caster-aware digital twin and motion-learning study on the actual Arche platform. I will present the engineering question, bounded semester deliverables and the specific support required. The immediate outcome is an access and experiment agreement with responsible people and dates.

Preferred date: Tuesday 6 October or Wednesday 7 October, for a 90-minute visit. If necessary, Tuesday 13 October or Wednesday 14 October is the fallback. All dates need confirmation. Participants: student and advisor, plus Arche's platform/software contact and the person responsible for mechanical information and safe robot access.

Minutes	Activity	Required result
0–5	Introductions and company use case	Which motion problem and platform matter most.
5–17	Student presentation	Shared scope, outputs and access requests.
17–35	Robot and caster inspection	Hardware revision, geometry, constraints and safe area.
35–55	CAD, software and sample data review	Available files, interfaces, channels and export route.
55–70	Experiment feasibility	Pose reference, operator, surfaces and session dates.
70–85	AI model and information permissions	Existing model access, data/code use and publication limits.
85–90	Decisions and ownership	Named contact, due dates, next slot and unresolved items.

Materials to bring

Bring the English visit slides, this timetable, the previous research PDF as a technical appendix, the access inventory and the recording checklist. Request a short sample log and available CAD/robot description before the meeting. Show implemented work only if it has been verified; diagrams and plans remain labelled as proposed.

Opening statement

"I am carrying out this ME492 project individually with my advisor's supervision. My goal is to model the caster interactions on your robot and test whether a smoother motion profile improves tracking and motion quality. I also plan a small learning experiment using the same recordings. Today I would like to agree on the platform information, data access and supervised test sessions so the work produces results that are useful to Arche and defensible academically."



Visit Access Inventory

Questions to answer with evidence owners and dates

For every item, record Available, Restricted, Unavailable or Pending, along with the owner, delivery date and scope of permission. “Available” requires a file, tested interface or confirmed booking. Keep these distinctions in the meeting record.

Item	Question and requested evidence
Actual platform	Which robot/hardware revision? Is differential drive the actual configuration? Request wheel/caster layout, mass/payload limits and identification photos if allowed.
CAD and mechanics	Can we receive CAD, simplified collision geometry, URDF/SDF, wheel radius/track, caster trail/position/material and mass/inertia data? What can be measured on site?
Software and commands	What OS/ROS version, drivers, APIs and controller interfaces exist? Is motion command replay permitted? Request interface documentation and a tested sample connection.
Telemetry	Which encoder, IMU, odometry, command, current and caster signals exist? At what rates and timestamps? Request a sample native log and channel description.
Independent reference	What external pose/video equipment exists? Can we use calibrated markers and a camera? What measurement accuracy and synchronization can we establish?
Robot access	Who supervises? Which two surfaces and area are permitted? Can Arche confirm the pilot, calibration and confirmation slots or offer operator-run capture?
Existing AI model	Does Arche have a relevant model? What task, inputs, outputs, weights, inference code and dataset access are available? Are training/adaptation and offline benchmarking allowed?
Compute and equipment	What student/lab/robot computer is available? What CPU, memory and GPU constraints apply? Who supplies permitted cameras/sensors and pays for consumables?
Data and ownership	What may leave the company? May logs/code enter external AI or cloud services? Who owns project code, derived datasets and trained weights? What may appear in the thesis/defense or public repository?
Publication and review	Is an NDA required? Who checks company information, within what agreed turnaround? Can we retain a reproducible academic package and a sanitized public subset?

Read-only access, exported data and supervised tests are sufficient for the initial study. Ask for the smallest working access route. The announcement’s estimated expense of 0 TL is not confirmation that additional sensors, travel or compute are funded. [4]



Visit Follow Through and Weekly Execution

Decisions actions and advisor visibility

Before and after the visit

By 2 October, prepare the deck and ask the advisor to coordinate the 6/7 October visit. Two working days before the confirmed meeting, provide the short agenda and request that Arche's technical contact attend. During the meeting, the student records decisions while the advisor confirms academic boundaries. No firm commitment is assumed without a named owner.

Within one working day after the visit, prepare a written record of platform revision, granted/restricted access, file and log delivery dates, test slots, instrumentation, publication rules, costs and open questions. The advisor and Arche contact confirm or correct the record. Within three working days, test the supplied files/sample log and update only assumptions affected by evidence. Retain the academic deadlines.

Action	Owner	Due or evidence
Visit coordination	Advisor with student support	Target 6/7 Oct, fallback 13/14 Oct.
CAD and interface sample	Arche named technical contact	Request within 3 working days after visit.
Usable logging pilot	Student and Arche operator	Target 20/21 Oct. G2 review 23 Oct.
Test and data permissions	Arche and advisor	By G1 16 Oct or recorded restricted route.
Academic reduced-scope decision	Advisor and student	By 9 Nov if G3 lacks physical evidence.

Weekly operating rhythm

Monday: select one demonstrable weekly output and check dependencies. Tuesday/Wednesday: prioritize confirmed visits or experiments; otherwise model. Friday: publish a short GitHub progress record with the planned output, actual work, evidence link, hours, blocker, next output and decision request. Target a 15–20 minute advisor review weekly, subject to availability. Distinguish prepared, executed and independently verified work. Publish company material only within confirmed permissions.

Advisor briefing - first three to four minutes

"I have prepared the semester execution baseline: scope, experiments, Arche access needs, decision gates and deadlines in one plan. It also states how the academic scope changes if company access is delayed. The 252 hours cover committed work; 32 hours are a risk reserve, not feature development."

Show page 1 for objectives/scope, pages 3–4 for the timetable, then page 5 for risk and fallback decisions. Explain that the remaining pages contain the experiment protocol, secondary learning task and Arche access checklist. Use the public GitHub project home for progress, evidence, decisions and current reports between meetings.



Sources and Status

References supporting the final execution baseline

[1] ME492 Fall 2026 course calendar, checked 30 September 2026. <https://yeditepe-me492.github.io/2026-Fall/>

[2] Yeditepe academic calendar, checked 30 September 2026. Classes end 25 December; undergraduate finals 26 December–8 January. <https://www.yeditepe.edu.tr/en/academic/academic-calendar>

[3] Existing local research reference: ME492_Caster_Digital_Twin_Research_Architecture_EN_2026-09-29.pdf, delivery folder. Reused technical scope and visual header; this schedule supersedes its provisional team/timeline assumptions.

[4] KKS_AUTOMR.docx, official Fall 2026 project announcement supplied locally. Individual execution reflects the newer advisor agreement reported by the student.

[5] Gazebo official ROS installation guidance, accessed 30 September 2026. Jazzy/Harmonic is the starting pair if compatible with Arche's platform; the visit determines the environment. https://gazebo.org/docs/latest/ros_installation/

Current baseline and progress

Revision 1.1 makes RQ1–RQ3 explicit, identifies motion learning as secondary, explains practical replication and adds the advisor briefing. Dates, 252 planned hours and 32 reserve hours are unchanged. Company access and experiment results remain pending; the public record distinguishes prepared, executed and verified work.

Public project home: <https://github.com/t0ttora/me492-caster-digital-twin>